AJKSA Community Platform

Feature Implementation Status: Frontend (citizen-frontend) and Backend (community-membership-management), measured against the Business Proposal

75	25	29	21
Total features audited	Implemented (green)	Partial (amber)	Not implemented (red)

How this was tested

(1) Read every controller in the backend (204 endpoints) and every page in the frontend (46 pages). (2) Called about 30 read-only endpoints on the running backend (localhost:8080) as the seeded super admin and compared each response with what the frontend expects. (3) Ran the TypeScript compiler on the frontend (clean). (4) Reviewed the existing backend test report: 102 tests, 0 failures. Maven is not installed here, so I could not re-run it. I did not create test users or applications, and I did not click through the UI in a browser. **Status key: Implemented** = works end to end in code. **Partial** = exists but incomplete, or shows sample data on error. **Not implemented** = missing, or present but broken. **Priority:** P0 blocks the core workflow, P1 high, P2 medium, P3 low.

Approval queue: root causes

- **The queue is empty because of a response-shape mismatch.** The backend returns `{data:{data:[...], totalCount}}`. The Approvals, Applications and Cards pages read `.content` and `totalElements`. Live, the backend holds 3 pending tasks (all routed to Super Admin) and the page shows none of them.
- **Fake data hides the failure.** If the call throws, the page shows invented tasks (APP-2026-8901) and counts (14 approved, 2 rejected). The summary also expects `pendingCount`, but the API sends `total_pending_tasks`.
- **No auto-submit.** Saving a complete profile does not send it for approval. The member has to find Submit on the Digital ID Card page.
- **Only one approver gets the task.** Routing picks Village Admin, else District Admin, else Super Admin. Your requirement is that it reaches all three levels. That needs a visibility change (a district or super admin sees everything under their scope) or three notifications. Currently only the first match is notified.
- **Village admins cannot open it.** The sidebar lists Approvals Queue for Super and District only. The /pending endpoint also ignores status, so approved items appear as 'pending'.
- **No 'return for correction' button.** The backend supports it. The frontend client and UI do not.

Feature matrix

Green = implemented, amber = partial, red = not implemented or broken.

1. Authentication & Accounts

#	Feature	Status	Priority	Target date	Evidence / notes
1	**User sign-up (username, email, password)**	**Implemented**	**P0**	**Done**	Register page + POST /auth/register. Email OTP confirms the account.
2	**OTP verification at sign-up**	**Implemented**	**P0**	**Done**	Works by email only. No SMS or WhatsApp provider is wired. Dev mode logs the OTP to the server log.
3	**Login (username/email + password, JWT)**	**Implemented**	**P0**	**Done**	Live test: super admin login returned 200 with access and refresh tokens.
4	**Token refresh and logout**	**Implemented**	**P0**	**Done**	Axios interceptor refreshes on 401 and redirects to /login if refresh fails.
5	**Login with OTP**	**Partial**	**P1**	**29 Oct 26**	Login page links to /verify-otp, but delivery is email only.
6	**Forgot / reset password**	**Implemented**	**P1**	**Done**	Pages and APIs exist. Not exercised live in this audit.
7	**Role-based landing page (super, district, village, member)**	**Implemented**	**P0**	**Done**	getDashboardRoute redirects by role.
8	**RBAC with geographic scope (global, district, village)**	**Implemented**	**P0**	**Done**	Live /me/scopes works. 16 backend scope and permission tests pass.
9	**Audit trail of admin actions**	**Partial**	**P1**	**29 Oct 26**	Backend AuditService writes events. The Audit Logs page shows hard-coded SAMPLE_AUDIT_LOGS and no API call, so real logs are not visible.

2. Geography & Administrators

#	Feature	Status	Priority	Target date	Evidence / notes
10	**Region, district, block, panchayat and village master data**	**Implemented**	**P0**	**Done**	Live: 2 regions and 20 districts seeded. Selector component used in forms.
11	**Tehsil and Area levels (in proposal)**	**Partial**	**P2**	**16 Nov 26**	Entities exist, but there is no API or UI for them.
12	**Super admin creates or edits regions, districts and villages**	**Not implemented**	**P1**	**2 Nov 26**	Geography API is read-only (GET only).
13	**Create and assign Village / District administrators**	**Implemented**	**P0**	**Done**	Administrators page + /admin/administrators. Live: 1 village admin assigned.
14	**User management (search, activate, deactivate, delete)**	**Implemented**	**P1**	**Done**	Admin Users page backed by /admin/users.
15	**Super admin dashboard**	**Implemented**	**P1**	**Done**	Live KPIs from /admin/dashboard/stats and /analytics/dashboard.
16	**District admin dashboard**	**Partial**	**P1**	**2 Nov 26**	Page is about 66 lines. A thin shell with no district KPIs, queue or reports.
17	**Village admin dashboard**	**Partial**	**P1**	**2 Nov 26**	Page is about 47 lines. A thin shell.

3. Member Profile & Registration

#	Feature	Status	Priority	Target date	Evidence / notes
18	**Member self-service profile (edit, save, photo upload)**	**Implemented**	**P0**	**Done**	732-line profile page with completion % from the server. Photo stored as base64 in the database.
19	**Admin registers a citizen (multi-step form)**	**Implemented**	**P0**	**Done**	/admin/citizens/new with geographic selector.
20	**Citizens directory (search, filter, view)**	**Implemented**	**P0**	**Done**	Live: /citizens returns data.
21	**Proposal fields: gender, marital status, street, area/tehsil, state, alternate number**	**Not implemented**	**P1**	**2 Nov 26**	Not on the Citizen entity.
22	**Proposal fields: voter ID, other IDs, skills, volunteer status**	**Not implemented**	**P1**	**2 Nov 26**	Not on the Citizen entity. Only Aadhaar is stored.
23	**Document upload (ID, certificates)**	**Not implemented**	**P1**	**5 Nov 26**	Only a profile photo can be uploaded. S3 bucket config exists, but no document API or UI.
24	**Members directory and member detail page**	**Partial**	**P1**	**29 Oct 26**	UI exists. The /members list is empty live because no member exists until an approval completes. The detail page falls back to sample data.
25	**Search by name, parent, mobile, village, area, house no.**	**Partial**	**P1**	**2 Nov 26**	Citizens search param and user search exist. No unified member search across all proposal fields.
26	**Membership types and renewal date handling**	**Partial**	**P2**	**16 Nov 26**	MembershipType entity exists. No UI to manage types or renewals.

4. Household & Family

#	Feature	Status	Priority	Target date	Evidence / notes
27	**Household and family CRUD (API)**	**Implemented**	**P0**	**Done**	Households and families controllers exist. Households list is empty live.
28	**Households admin page**	**Partial**	**P1**	**5 Nov 26**	UI is wired, but falls back to sample data when the API fails.
29	**Families & Family Tree page**	**Not implemented**	**P1**	**5 Nov 26**	Page uses hard-coded sample families (FAM-2026-9901) and makes no API calls.
30	**Join an existing family by OTP (family-join)**	**Not implemented**	**P1**	**5 Nov 26**	Backend /family-join APIs exist. No frontend page calls them.
31	**Household database fields (GPS, students, seniors, photo)**	**Not implemented**	**P2**	**16 Nov 26**	Not modelled.

5. Membership Approval Workflow (key gap raised)

#	Feature	Status	Priority	Target date	Evidence / notes
32	**Backend: submit, 3-tier routing, tasks, history, approve / reject / return**	**Implemented**	**P0**	**Done**	Routes Village Admin, then District Admin, then Super Admin. Approval creates the Member and issues the card. Live: 3 applications and 3 pending tasks exist.

5. Membership Approval Workflow (key gap raised)

#	Feature	Status	Priority	Target date	Evidence / notes
33	**Approval Queue page (/admin/approvals)**	**Not implemented**	**P0**	**27 Oct 26**	BROKEN. It reads response.content, but the API returns data.data, so the queue is always empty even with 3 pending tasks. Summary keys also do not match. On errors it shows fake tasks and counts (14 approved, 2 rejected).
34	**Applications list page (/admin/applications)**	**Not implemented**	**P0**	**27 Oct 26**	Same defect: reads .content, so real applications never appear. Sends page/size, but the API expects pageNo/pageSize.
35	**Profile completed -> goes to village, district and super admin**	**Partial**	**P0**	**27 Oct 26**	Not automatic. The member must press Submit on the Digital ID Card page. The backend creates one task for one approver only (first match) and not three.
36	**Village admin can see the queue**	**Not implemented**	**P0**	**27 Oct 26**	Sidebar hides 'Approvals Queue' from VILLAGE_ADMIN. The backend allows it.
37	**District / Super admin see applications in their scope**	**Partial**	**P0**	**27 Oct 26**	The /pending endpoint passes no status filter, so it returns all statuses. The UI cannot display the result because of the .content mismatch.
38	**Return for correction / request changes (admin UI)**	**Not implemented**	**P0**	**27 Oct 26**	Backend /return exists. The frontend API client has no method and the UI has no button.
39	**Applicant sees status, correction reason, can resubmit**	**Implemented**	**P0**	**Done**	CardAccessGate shows each state and has resubmit.
40	**Card unlocks only after approval**	**Implemented**	**P0**	**Done**	Server-calculated card_eligible gates /member/card.
41	**Notify approver and applicant**	**Partial**	**P1**	**29 Oct 26**	Backend creates in-app notifications. No visible notification centre in the UI.

6. Digital Membership Card & Verification

#	Feature	Status	Priority	Target date	Evidence / notes
42	**Digital card (front and back, photo, QR)**	**Implemented**	**P0**	**Done**	DigitalMembershipCard + qrcode.react. Printable.
43	**Public QR verification page**	**Partial**	**P0**	**27 Oct 26**	Page and POST /cards/verify exist. On error the page shows a demonstration result instead of failing, so a failed lookup could look valid.
44	**Admin card list, issue, revoke**	**Partial**	**P1**	**29 Oct 26**	Reads .content, so the list is empty. Falls back to sample data.

7. Communication

#	Feature	Status	Priority	Target date	Evidence / notes
45	**Targeted broadcast (district, village, role)**	**Partial**	**P1**	**9 Nov 26**	Backend and UI exist. Proposal targets such as tehsil, women, youth and seniors are not available.
46	**In-app notifications**	**Partial**	**P1**	**29 Oct 26**	Backend stores them. No member-side inbox page.
47	**SMS / WhatsApp / Push (FCM) / Email delivery**	**Not implemented**	**P1**	**12 Nov 26**	UI offers SMS options. Backend has no SMS, WhatsApp or FCM provider. Only OTP mail via JavaMailSender.
48	**Outbox monitoring**	**Partial**	**P2**	**16 Nov 26**	Live outbox has pending events. The page keeps a mock fallback.

7. Communication

#	Feature	Status	Priority	Target date	Evidence / notes
49	**Community news feed, notices, circulars with media**	**Not implemented**	**P2**	**16 Nov 26**	No module in backend or frontend.

8. Events, Meetings & Volunteers

#	Feature	Status	Priority	Target date	Evidence / notes
50	**Meetings: create, RSVP, geofenced check-in, report**	**Implemented**	**P1**	**Done**	Backend /meetings full set. Admin and member pages exist. Member events page falls back to sample meetings.
51	**Events (create, register)**	**Partial**	**P1**	**9 Nov 26**	Backend /events exists. No dedicated frontend. Member Events screen uses meetings.
52	**QR event attendance**	**Partial**	**P2**	**16 Nov 26**	Geo check-in exists. No QR scan flow.
53	**Campaigns**	**Partial**	**P2**	**16 Nov 26**	Backend and page exist. Page falls back to sample data.
54	**Volunteer management (skills, availability, blood donor)**	**Not implemented**	**P2**	**20 Nov 26**	No module.

9. Emergency & Safety

#	Feature	Status	Priority	Target date	Evidence / notes
55	**SOS button, I'm Safe check-in, emergency contact alerts**	**Not implemented**	**P2**	**20 Nov 26**	Not in backend or frontend.
56	**Consent-based temporary location sharing**	**Not implemented**	**P2**	**20 Nov 26**	Not implemented.

10. Governance & Community Services

#	Feature	Status	Priority	Target date	Evidence / notes
57	**Grievances (raise, assign, resolve)**	**Partial**	**P1**	**9 Nov 26**	Backend complete. Both pages fall back to sample data on error.
58	**Polls and voting**	**Partial**	**P2**	**16 Nov 26**	Backend complete. Pages fall back to sample polls and 'optimistic' votes.
59	**Life events (birth, death, marriage) and verification**	**Partial**	**P2**	**16 Nov 26**	Backend and pages exist. Admin page falls back to sample data.
60	**Committees and member assignment**	**Partial**	**P2**	**16 Nov 26**	Backend complete. Page falls back to sample committees.

11. Finance

#	Feature	Status	Priority	Target date	Evidence / notes
61	**Subscriptions, payments, donations, financial summary (API)**	**Implemented**	**P1**	**Done**	Live: /finance/summary, /subscriptions, /donations return 200.
62	**Finance pages**	**Partial**	**P1**	**9 Nov 26**	UI exists. Fallback sample metrics. Payments page uses an 'optimistic fallback for demo'.
63	**Online payment gateway**	**Not implemented**	**P2**	**12 Nov 26**	Payments are marked complete or failed by API call. No gateway integration.

12. Reports, Analytics & Data

#	Feature	Status	Priority	Target date	Evidence / notes
64	**Executive analytics dashboard**	**Implemented**	**P1**	**Done**	Live KPIs via /analytics/dashboard.
65	**Population, occupation and health analytics (proposal)**	**Not implemented**	**P2**	**20 Nov 26**	Not in the API. Dashboard charts use demo data (flagged 'Demo data').
66	**Report generation by village, district, gender, age**	**Partial**	**P1**	**9 Nov 26**	Reports API exists (empty live). No reports UI.
67	**Export to CSV / Excel / PDF**	**Partial**	**P1**	**9 Nov 26**	Export jobs create CSV file names only. No Excel or PDF.
68	**Bulk import**	**Partial**	**P2**	**16 Nov 26**	Import job API and UI exist. The UI shows a simulated job ('498 records') on error.
69	**System settings**	**Partial**	**P3**	**25 Nov 26**	Page and API exist. No settings stored.

13. Platform / Non-functional

#	Feature	Status	Priority	Target date	Evidence / notes
70	**Payload encryption (AES-GCM)**	**Implemented**	**P1**	**Done**	Backend filter, key provider and tests pass. Frontend does not use it.
71	**Backend automated tests**	**Implemented**	**P1**	**Done**	102 tests across 26 classes, 0 failures (report dated 19 Oct 2026). I could not re-run: Maven is not installed here.
72	**Frontend type-check**	**Implemented**	**P1**	**Done**	tsc --noEmit passes with no errors. There are no frontend tests; the e2e script references Playwright, which is not installed.
73	**Multi-language support**	**Not implemented**	**P3**	**25 Nov 26**	No i18n.
74	**Mobile apps, offline sync, AI search**	**Not implemented**	**P3**	**15 Dec 26**	Roadmap items. Not started.
75	**Daily automated backups**	**Not implemented**	**P2**	**16 Nov 26**	No evidence in the repo.

Timeline and recommended order of work

Features are built with AI assistance in batches, highest priority first, each batch about 2 to 4 days after the previous one. Dates are targets. SMS/WhatsApp and the payment gateway also depend on the client providing provider accounts and credentials.

1. **Batch 1, by 27 Oct:** approval workflow end to end and QR verify (all P0). **Batch 2, 29 Oct:** audit logs, members, cards, notifications, OTP login. **Batch 3, 2 Nov:** profile fields, geography admin, dashboards, search. **Batch 4, 5 Nov:** documents, households, families, family-join. **Batch 5, 9 Nov:** broadcast, events, grievances, finance, reports, export. **Batch 6, 12 Nov:** SMS/WhatsApp and payment gateway (need provider accounts). **Batch 7, 16 Nov:** remaining P2 modules. **Batch 8, 20 Nov:** volunteers, SOS, analytics. **Batch 9, 25 Nov:** settings and i18n. **Mobile apps, by 15 Dec.**
2. **Fix the list contract (P0, about half a day).** Map the backend `data / totalCount / page` shape in the API client once, and send `pageNo / pageSize`. This unblocks the Approvals, Applications and Cards pages.
3. **Remove silent fallbacks (P0).** Replace the sample-data catch blocks with real error states. Start with approvals, applications, QR verify and finance.
4. **Finish the approval queue (P0).** Add Return for Correction, show the queue to Village Admin, filter /pending by active statuses, and make District and Super Admin see all applications in their scope.
5. **Auto-submit on profile completion (P0).** Either call submit when the profile reaches 100%, or add a Submit for Approval button on the profile page. Notify all three tiers.
6. **Close the proposal data gaps (P1).** Add gender, marital status, voter ID, skills, volunteer status, tehsil and area; add document upload; wire Families and Family-Join to the API.
7. **Real delivery channels (P1).** Add SMS or WhatsApp for OTP and broadcasts. Then reports export (Excel and PDF) and a real audit-log page.
8. **Later (P2/P3).** Volunteers, SOS, news feed, payment gateway, backups, i18n, mobile apps.